

RMAN

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Configuring Your Database for Basic Backup and Recovery

- Use a fast recovery area, which automates storage management for most backup-related files, and specify it as an archived redo log file destination.
- Run the database in ARCHIVELOG mode so you can perform online backups and have data recovery options such as complete and point-in-time media recovery.

Starting and Exiting RMAN

- Start RMAN without specifying any connection options
% rman
RMAN> CONNECT TARGET /
OR
RMAN> CONNECT TARGET SYS/oracle@trgt
- Start the RMAN with target database option
% rman TARGET /
% rman TARGET SYS/oracle@trgt
- To quit RMAN and terminate the program, type EXIT or QUIT at the RMAN prompt
RMAN> EXIT

Setting Globalization Support Environment Variables for RMAN

- Before invoking RMAN, set the NLS_DATE_FORMAT and NLS_LANG environment variables
 - NLS_LANG=american
 - NLS_DATE_FORMAT='Mon DD YYYY HH24:MI:SS'

RMAN Configuration

- RMAN> SHOW ALL;
 - shows all CONFIGURE settings, both user-entered and default
- CONFIGURE... CLEAR
 - Restoring Default RMAN Configuration Settings
 - E.g.
 - RMAN>CONFIGURE BACKUP OPTIMIZATION CLEAR;
- Configuring the Default Device Type for Backups
 - RMAN>CONFIGURE DEFAULT DEVICE TYPE TO DISK
- Configuring Control File and Server Parameter File Autobackup
 - CONFIGURE CONTROLFILE AUTOBACKUP ON;
- Configuring the Backup Retention Policy
 - RMAN> CONFIGURE RETENTION POLICY TO RECOVERY WINDOW OF 7 DAYS; //This example ensures that you can recover the database to any point within the last week
 - RMAN> CONFIGURE RETENTION POLICY TO REDUNDANCY 3; //This ensures that 3 copies of backups will be stored
 - RMAN> SHOW RETENTION POLICY;

Setting Up a Flash Recovery Area for RMAN

- flash recovery area simplifies the ongoing administration of your database by automatically naming files, retaining them as long as they are needed for restore and recovery activities, and deleting them when they are no longer needed to restore your database and space is needed for some other backup and recovery-related purpose.
- **Setting Initialization Parameters for the Flash Recovery Area**
 - DB_RECOVERY_FILE_DEST_SIZE : which specifies the disk quota, or maximum space to use for flash recovery area files for this database
 - and DB_RECOVERY_FILE_DEST : which specifies the location of the flash recovery area
 - SQL>ALTER SYSTEM SET DB_RECOVERY_FILE_DEST_SIZE=100G SCOPE=BOTH;
 - SQL>ALTER SYSTEM SET DB_RECOVERY_FILE_DEST='/u01/flash_recovery_area' SCOPE=BOTH;

Note:- DB_RECOVERY_FILE_DEST_SIZE must be set *before* DB_RECOVERY_FILE_DEST.

```
SQL> SELECT * FROM V$RECOVERY_FILE_DEST;
```

- **MOUNT ARCHIVE LOG DESTINATION**
 - SQL>ALTER SYSTEM SET LOG_ARCHIVE_DEST_1='LOCATION=/u02/archive_logfile' SCOPE=spfile;
 - SQL>ALTER SYSTEM SET LOG_ARCHIVE_DEST_10='LOCATION=USE_DB_RECOVERY_FILE_DEST' SCOPE=spfile;
 - SQL>alter system set log_archive_format='arch_%t_%s_%r.arc' SCOPE=spfile;

Datatabase Archiving

- SQL>shutdown immediate;
- SQL>startup mount;
- SQL>alter database archivelog;
- SQL>alter database open;
- SQL>ALTER SYSTEM ARCHIVE LOG START;
- SQL>ALTER SYSTEM SWITCH LOGFILE;

Making Backups with Recovery Manager

- RMAN backups can be taken when the database is mounted or open.
RMAN> BACKUP DATABASE;
- Files That RMAN Can Back Up
 - Datafiles, and image copies of datafiles
 - Control files, and image copies of control files
 - Archived redo logs
 - The current server parameter file
 - Backup pieces, containing other backups created by RMAN
- network configuration files, password files, and the contents of the Oracle home, these files cannot be backed up with RMAN. Use some non-RMAN backup solution for any files not in the preceding list.
- RMAN backups can be stored in one of two formats: as **image copies** or as **backup sets**.
- **Image Copies**
 - An image copy is a bit-for-bit duplicate of a database file, identical to a copy made with an operating system command
 - RMAN creates image copies when the AS COPY option is used with the BACKUP command
- **Backup Sets**
 - backup information in logical structures
 - A backup set contains the data from one or more datafiles or archived redo logs, or control files or SPFILE
 - Datafiles and archivelogs cannot be mixed together in the same backup set
 - A backup set consists of one or more files called **backup pieces**, files in an RMAN-specific format.

Backing Up Database Files and Archived Logs with RMANs

- use the BACKUP DATABASE command.
 - RMAN> BACKUP DATABASE; # uses automatic channels to make backup
 - RMAN> SQL 'ALTER SYSTEM ARCHIVE LOG CURRENT'; # switches logs and archives all logs
- By archiving the logs immediately after the backup, you ensure that you have a full set of archived logs through the time of the backup. This guarantees that you can perform media recovery after restoring this backup.
- The FORMAT parameter to the BACKUP DATABASE command lets you specify a different destination for backups and control the filename for the backup.
 - RMAN> BACKUP DATABASE FORMAT '/tmp/%U'; # %U generates a unique filename
- You can also use the FORMAT argument to name an ASM disk group as backup destination, as shown in this example:
 - RMAN> BACKUP DATABASE FORMAT '+dgroup1'; # sets an ASM disk group

- **To back up a tablespace:**

RMAN>BACKUP DEVICE TYPE sbt MAXSETSIZE = 10M TABLESPACE
users, tools;

- **To back up a datafile:**

RMAN>BACKUP DEVICE TYPE sbt
DATAFILE 1,2,3,4
DATAFILECOPY '/tmp/system01.dbf';

This example uses an sbt channel to back up datafiles 1-4 as well as a datafile copy

- If CONFIGURE CONTROLFILE AUTOBACKUP is ON, then RMAN writes the current control file and SPFILE to a separate autobackup piece. Otherwise, these files are automatically included in the backup set that contains datafile 1.

- **Backing Up Datafile Copies**

- BACKUP DEVICE TYPE disk DATAFILECOPY '/tmp/system01.dbf';

Backing Up Control Files with RMAN

- If CONFIGURE CONTROLFILE AUTOBACKUP is ON (by default it is OFF), then RMAN automatically backs up the control file and server parameter file after every backup and after database structural changes.
- The control file autobackup contains metadata about the previous backup, which is crucial for disaster recovery.
- If the autobackup feature is not set, then you must manually back up the control file in one of the following ways:
 - Run BACKUP CURRENT CONTROLFILE
 - Include a backup of the control file within any backup by using the INCLUDE CURRENT CONTROLFILE option of the BACKUP command
 - Back up datafile 1, because RMAN automatically includes the control file and SPFILE in backups of datafile 1

RMAN> backup current controlfile;

RMAN> backup datafile 4 include current controlfile;

- A manual backup of the control file is not the same as a control file autobackup.
- To create an image copy of the control file, use the following command:
 - *RMAN> backup as copy current controlfile format c:\bcp_controlfile.ctl;*
- **Backing Up Server Parameter Files with RMAN**
 - BACKUP DEVICE TYPE sbt SPFILE;
 - The SPFILE that is backed up is the one currently in use by the instance.

Backing Up Archived Redo Logs with RMAN

- Archived redo logs are the key to successful media recovery.
- You can back up logs with `BACKUP ARCHIVELOG`, or back up logs while backing up datafiles and control files by specifying `BACKUP DATABASE PLUS ARCHIVELOG` command.
- **Backing Up Logs with `BACKUP ... PLUS ARCHIVELOG`**
 - Adding `BACKUP ... PLUS ARCHIVELOG` causes RMAN to do the following:
 - Runs the `ALTER SYSTEM ARCHIVE LOG CURRENT` command.
 - Runs `BACKUP ARCHIVELOG ALL`. Note that if backup optimization is enabled, then RMAN skips logs that it has already backed up to the specified device.
 - Backs up the rest of the files specified in `BACKUP` command.
 - Runs the `ALTER SYSTEM ARCHIVE LOG CURRENT` command.
 - Backs up any remaining archived logs generated during the backup.
 - This guarantees that datafile backups taken during the command are recoverable to a consistent state.

Database Restore and Recovery with RMAN

- The database files that RMAN backs up and can recover include the control file, server parameter file, datafiles and archived redo log files
- The two most important RMAN commands used in database recovery are:
 - RESTORE, which retrieves files from RMAN backups based on the contents of the RMAN repository
 - RECOVER, which performs complete or point-in-time media recovery using available datafiles and redo logs.
- **Using RESTORE... PREVIEW**
 - RESTORE ... PREVIEW can be applied to any RESTORE operation to create a detailed report of every backup to be used in the requested RESTORE operation
 - RESTORE DATABASE PREVIEW;
 - RESTORE TABLESPACE users PREVIEW;
 - RESTORE DATAFILE 3 PREVIEW;
 - RESTORE ARCHIVELOG FROM LOGSEQ 200 PREVIEW;
- **Validating with RESTORE ... VALIDATE**
 - The RESTORE ... VALIDATE and VALIDATE BACKUPSET commands test whether you can restore from your backups
 - You do not have to take datafiles offline when validating them
 - RESTORE CONTROLFILE VALIDATE;
 - RESTORE TABLESPACE SYSTEM VALIDATE;
 - RESTORE ARCHIVELOG ALL VALIDATE;
 - If you do not see an error stack, then RMAN successfully tested restore of the specified objects from the available backups.

Whole Database Restore and Recovery: Scenario

- In this scenario, you have a current control file and SPFILE but all datafiles are damaged or lost. You must restore and recover the whole database.
 1. After connecting to the target database, make sure the database is mounted.
RMAN> STARTUP MOUNT
 2. Use the SHOW ALL command to see what channels are configured for access to backup devices. If automatic channels are not configured, then manually allocate one or more channels.
 3. Restore the database using the RESTORE command, and recover it using the RECOVER command.
 4. Examine the output to see if recovery was successful. If so, open the database.

The following example performs restore and recovery of the database, using automatic channels.

```
RMAN> RESTORE DATABASE;  
RMAN> RECOVER DATABASE DELETE ARCHIVELOG MAXSIZE 25MB;
```
- The RECOVER DATABASE command as used here illustrates two useful options:
 - DELETE ARCHIVELOG causes RMAN to delete restored log files after they have been applied to the datafiles, to save disk space.
 - MAXSIZE 25MB limits space occupied by restored logs at any given moment to 25MB. This gives you more control over disk space usage by the restored logs. Note that if a single archived redo log file is larger than the specified MAXSIZE value, you will get an error. You will have to try your command again with a larger MAXSIZE value.
- You can force RMAN to restore any missing datafiles belonging to read-only tablespaces by using the CHECK READONLY option to the RESTORE command:
 - RMAN> RESTORE DATABASE CHECK READONLY;
 - RMAN> RECOVER DATABASE DELETE ARCHIVELOG;
- If RMAN completes the recovery without error, you can open the database:
 - RMAN> ALTER DATABASE OPEN;

Restore and Recovery of Individual Tablespaces or Datafiles: Scenario

- In this scenario, some but not all of the datafiles are damaged. You want to leave the database open so that the undamaged datafiles remain available.
- The following restore and recovery procedure can be used if the database is mounted or open.
 1. Connect to the target database and the recovery catalog database (if applicable), and make sure the database is mounted or open.
 2. Take the tablespaces affected offline using `ALTER TABLESPACE ... OFFLINE IMMEDIATE` if they are not already offline.
 3. Run `SHOW ALL` to see the current configuration, including configured default channels for the device where the backup is stored. If necessary, allocate one or more channels manually. (Note that if you manually allocate channels you must use a `RUN` block around your command.)
 4. Restore the tablespace or datafile with the `RESTORE` command, and recover it with the `RECOVER` command.
- This example restores and recovers the users tablespace, letting RMAN choose the backup to use on disk or tape:
 - `RMAN> SQL 'ALTER TABLESPACE users OFFLINE IMMEDIATE';`
 - `RMAN> RESTORE TABLESPACE users;`
 - `RMAN> RECOVER TABLESPACE users;`
- If RMAN reported no errors during the recovery, then bring the tablespace back online:
 - `RMAN> SQL 'ALTER TABLESPACE users ONLINE';`

Restoring the Control File from Backup

- Loss or corruption of all copies of your control file requires restore of the control file from backup.
- The RESTORE CONTROLFILE command is used to restore the control file.
- RMAN can restore it to its default location (determined by rules described in the following section) or to one or more different locations of your choice, using the RESTORE CONTROLFILE... TO destination option.
- If you want to restore the control file from autobackup, the database must be in a NOMOUNT state
- You must first set the DBID for your database, and then use the RESTORE CONTROLFILE FROM AUTOBACKUP command:

```
RMAN> SET DBID 320066378;  
RMAN> RUN {  
    SET CONTROLFILE AUTOBACKUP FORMAT  
    FOR DEVICE TYPE DISK TO 'autobackup_format';  
    RESTORE CONTROLFILE FROM AUTOBACKUP;  
}
```


To restore the server parameter file:

- If the database is up at the time of the loss of the SPFILE, connect to the target database.
 - For example, run: `rman TARGET /`

If the database is not up when the SPFILE is lost, and you are not using a recovery catalog, then you must set the DBID of the target database.
- Shut down the instance and restart it without mounting. When the SPFILE is not available, RMAN starts the instance with a dummy parameter file. For example:
 - `RMAN> STARTUP FORCE NOMOUNT;`
- Restore the server parameter file. If restoring to the default location, then run:
 - `RMAN> RESTORE SPFILE FROM AUTOBACKUP;`

If restoring to a nondefault location, then you could run commands as in the following example:

 - `RMAN> RESTORE SPFILE TO '/tmp/spfileTEMP.ora' FROM AUTOBACKUP;`
- Restart the instance with the restored file. If restarting with a server parameter file in a nondefault location, then create a new client-side initialization parameter file with the single line `SPFILE=new_location`, where `new_location` is the path name of the restored server parameter file. Then, restart the instance with the client-side initialization parameter file.
 - For example, create a file `/tmp/init.ora` which contains the single line:
 - `SPFILE=/tmp/spfileTEMP.ora`
 - Then use this RMAN command, to restart the instance based on the restored SPFILE:
 - `RMAN> STARTUP FORCE PFILE=/tmp/init.ora; # startup with /tmp/spfileTEMP.ora`
- If you want to restore the SPFILE from the autobackup, you must first set the DBID for your database, and then use the `RESTORE SPFILE FROM AUTOBACKUP` command.

```
RMAN> SET DBID 320066378;
RMAN> RUN {
SET CONTROLFILE AUTOBACKUP FORMAT
FOR DEVICE TYPE DISK TO 'autobackup_format';
RESTORE SPFILE FROM AUTOBACKUP;
}
```

Using DataPump Export and Import Utility

- The Data Pump Export and Import utilities, first released in Oracle Database 10g Release 1 (10.1), are high performance replacements for the original Export and Import utilities.
- Data Pump Export utility is used to perform logical backup
- **Table Exports/Imports**
 - *CONN / AS SYSDBA*
 - *ALTER USER scott IDENTIFIED BY tiger ACCOUNT UNLOCK;*
 -
 - *CREATE OR REPLACE DIRECTORY test_dir AS '/u01/app/oracle/oradata/';*
 - *GRANT READ, WRITE ON DIRECTORY test_dir TO scott;*
 - *Existing directories can be queried using the ALL_DIRECTORIES view.*
- The TABLES parameter is used to specify the tables that are to be exported. The following is an example of the table export and import syntax.
 - *expdp scott/tiger@db10g tables=EMP,DEPT directory=TEST_DIR dumpfile=EMP_DEPT.dmp logfile=expdpEMP_DEPT.log*
 - *impdp scott/tiger@db10g tables=EMP,DEPT directory=TEST_DIR dumpfile=EMP_DEPT.dmp logfile=impdpEMP_DEPT.log*

DataPump Export and Import Utility...

- **Schema Exports/Imports**

- *expdp scott/tiger@db10g schemas=SCOTT directory=TEST_DIR dumpfile=SCOTT.dmp logfile=expdpSCOTT.log*
- *impdp scott/tiger@db10g schemas=SCOTT directory=TEST_DIR dumpfile=SCOTT.dmp logfile=impdpSCOTT.log*

- **Database Exports/Imports**

- The FULL parameter indicates that a complete database export is required. The following is an example of the full database export and import syntax.
 - *expdp system/password@db10g full=Y directory=TEST_DIR dumpfile=DB10G.dmp logfile=expdpDB10G.log*
 - *impdp system/password@db10g full=Y directory=TEST_DIR dumpfile=DB10G.dmp logfile=impdpDB10G.log*
- *directory=DATA_PUMP_DIR* option can be used for default directory
- *table_exists_action=replace* and *DIR table_exists_action=truncate* can be used on *impdp* utility as per need

ASM(Autonomous Storage Management)

- One of the most significant new features in Oracle 10g release is the introduction of Automated Storage Management (ASM)
- Oracle ASM is a volume manager and a file system for Oracle database files that supports single-instance Oracle Database and Oracle Real Application Clusters (Oracle RAC) configurations
- Oracle ASM uses disk groups to store data files; an Oracle ASM disk group is a collection of disks that Oracle ASM manages as a unit. The performance is comparable to the performance of raw devices.
- You can add or remove disks from a disk group while a database continues to access files from the disk group. When you add or remove disks from a disk group, Oracle ASM automatically redistributes the file contents and eliminates the need for downtime when redistributing the content.
- Oracle ASM also uses the Oracle Managed Files (OMF) feature to simplify database file management.
- Oracle ASM reduces the administrative overhead for managing database storage by consolidating data storage into a small number of disk groups.

Understanding Oracle ASM Concepts

- **Oracle ASM Instances**

- built on the same technology as an Oracle Database instance
- An Oracle ASM instance has a System Global Area (SGA) and background processes that are similar to those of Oracle Database. However, because Oracle ASM performs fewer tasks than a database, an Oracle ASM SGA is much smaller than a database SGA
- Oracle ASM instances mount disk groups to make Oracle ASM files available to database instances
- Oracle ASM instances do not mount databases.
- Oracle ASM is installed in the Oracle Grid Infrastructure home before Oracle Database is installed in a separate Oracle home.
- Oracle ASM and database instances require shared access to the disks in a disk group.
- Oracle ASM instances manage the metadata of the disk group and provide file layout information to the database instances.
- Oracle ASM metadata includes the following information:
 - The disks that belong to a disk group
 - The amount of space that is available in a disk group
 - The filenames of the files in a disk group
 - The location of disk group data file extents
 - A redo log that records information about atomically changing metadata blocks
 - Oracle ADVM volume information

- **Oracle ASM Disk Groups**

- A disk group is basically one or more ASM disks that are managed as a single logical unit.
- Each disk group contains the metadata that is required for the management of space in the disk group.
- Disk group components include disks, files, and allocation units.